

Lec15

Question Practise

Convert Names to Uppercase

Question 1

Given:

```
const names = ["gagan", "ishan", "yogesh"];
```

Return:

```
["GAGAN", "ISHAN", "YOGESH"]
```

Answer

```
const result = names.map(name => name.toUpperCase());  
console.log(result);
```

Extract Product Names

Given:

```
const products = [  
  { id: 1, name: "Laptop" },  
  { id: 2, name: "Phone" },  
  { id: 3, name: "Tablet" }  
];
```

Return:

```
["Laptop", "Phone", "Tablet"]
```

Answer

```
const result = products.map(product => product.name);  
console.log(result);
```

Filter Long Words

Question

Keep words having length greater than 5.

```
const words = ["apple", "banana", "cat", "elephant", "dog"];
```

Return:

```
["banana", "elephant"];
```

Answer

```
const result = words.filter(word => word.length > 5);  
console.log(result);
```

Filter Adults

Keep only users aged 18 or above.

```
const users = [  
  { name: "A", age: 16 },  
  { name: "B", age: 22 },  
  { name: "C", age: 19 }  
];
```


Answer

```
const result = users.filter(user => user.age >= 18);  
console.log(result);
```

Count Total Characters

Question

Find total characters in all strings.

```
const words = ["hello", "world", "javascript"];
```

Answer

```
const result = words.reduce(  
  (total, word) => total + word.length, 0);  
console.log(result);
```

Get Squares of Even Numbers

Given:

```
const nums = [1,2,3,4,5,6,7,8];
```

Return squares of only even numbers.

Answer

```
const result = nums  
  .filter(num => num % 2 === 0)  
  .map(num => num * num);  
console.log(result);
```

Total Salary of Employees

Question

```
const employees = [  
  { name: "A", salary: 30000 },  
  { name: "B", salary: 50000 },  
  { name: "C", salary: 40000 }  
];
```

Find total salary.

Answer

```
const result = employees.reduce(  
  (total, emp) => total + emp.salary, 0);  
console.log(result);
```

Names Of Passed Student

Question

Pass marks = 40

```
const students = [  
  { name: "A", marks: 35 },  
  { name: "B", marks: 75 },  
  { name: "C", marks: 45 },  
  { name: "D", marks: 25 }  
];
```

Return:

```
["B", "C"]
```


Answer

```
const result = students  
  .filter(student => student.marks >= 40)  
  .map(student => student.name);  
console.log(result);
```

Average Marks of Students

Question

`const marks = [80, 90, 70, 60];`

Find average.

Answer

```
const result =  
marks.reduce((sum, mark) => sum + mark, 0) / marks.length;  
console.log(result);
```

Count Occurrences of Elements

Question

```
const fruits = [  
  "apple",  
  "banana",  
  "apple",  
  "orange",  
  "banana",  
  "apple"  
];
```

Return:

```
{  
  apple: 3,  
  banana: 2,  
  orange: 1  
}
```

Answer

```
const result = fruits.reduce((acc, fruit) => {  
  acc[fruit] = (acc[fruit] || 0) + 1;  
  return acc;  
}, {});  
console.log(result);
```

Top Performing Students

```
const students = [  
  {  
    name: "Utkarsh",  
    marks: [80, 90, 85]  
  },  
  {  
    name: "Yukta",  
    marks: [95, 92, 98]  
  },  
  {  
    name: "Navneet",  
    marks: [60, 70, 65]  
  }  
];
```

Tasks:

- Calculate average marks
- Keep average ≥ 85
- Return names sorted by average descending

Answer

```
const result = students
  .map(student => ({
    name: student.name,
    avg:
      student.marks.reduce(
        (sum, mark) => sum + mark,
        0) / student.marks.length
  }))
  .filter(student => student.avg >= 85)
  .sort((a, b) => b.avg - a.avg)
  .map(student => student.name);

console.log(result);
```

Group Products by Category

Question

```
const products = [  
  { name: "Laptop", category: "Electronics" },  
  { name: "Phone", category: "Electronics" },  
  { name: "Shirt", category: "Clothing" }  
];
```

Return:

```
{  
  Electronics: ["Laptop", "Phone"],  
  Clothing: ["Shirt"]  
}
```


Answer

```
const result = products.reduce((acc, product) => {  
  
  if (!acc[product.category]) {  
    acc[product.category] = [];  
  }  
  
  acc[product.category].push(product.name);  
  
  return acc;  
  
}, {});  
  
console.log(result);
```

Total Cart Value

Question

```
const cart = [  
  { item: "Laptop", price: 50000, quantity: 1 },  
  { item: "Mouse", price: 500, quantity: 2 },  
  { item: "Keyboard", price: 1500, quantity: 1 }  
];
```

Find total cart value.

Answer

```
const result = cart.reduce(  
  (total, item) =>  
    total + item.price * item.quantity, 0);  
console.log(result);
```

Highest Scoring Student

Question

```
const students = [  
  { name: "Qureshi", marks: 75 },  
  { name: "Smiti", marks: 92 },  
  { name: "Sahil", marks: 88 }  
];
```

Return:

```
{  
  name: "Smiti",  
  marks: 92  
}
```

Answer

```
const result = students.reduce(  
  (highest, student) =>  
    student.marks > highest.marks ? student : highest);  
console.log(result);
```

Generate User Emails

Question

```
const users = [  
  "a",  
  "b",  
  "c"  
];
```

Return:

```
[  
  "a@gmail.com",  
  "b@gmail.com",  
  "c@gmail.com"  
]
```

Answer

```
const result = users.map(  
  user => `${user}@gmail.com`  
);  
  
console.log(result);
```